

Cognitive Misalignment Is Not Singular

— Redefining D2 as a Multi-Layer Misalignment Model —

0. Overview

The starting point of this paper lies in the continued observation of a difficult-to-classify sense of “something is correct, yet wrong” emerging during dialogue. Normally, this kind of discomfort disappears before it can be recorded or analyzed. The reason this observation became possible was the prior construction of a cognitive structure map known as SCOPE (Sephrotic Cognitive Process Engine). Because SCOPE existed as a coordinate system, it became possible for the first time to identify these discomforts as “misalignments occurring at specific layers.” ![[scopeシヨブ20260510214426_0001.png]]

※ Each layer of SCOPE (L0–L8) functions as a coordinate axis indicating where cognitive functions originate. For details, refer to the related PDFs.

The misalignment discussed in this paper (hereafter referred to as D2) is a provisional term referring to “premise mismatches” and “interpretive friction” occurring during dialogue and cognitive processing. Since this work is based on N=1 observations, and since the author is not a specialist simultaneously spanning all related disciplines involved, verification and definitive claims remain difficult. Therefore, this paper should be treated as a hypothesis until a verifiable environment becomes available.

In previous related papers, dialogue misalignment (D2) was treated as a singular form of “premise mismatch.”

However, in practice, unclassified discomfort and difficult-to-explain forms of misalignment existed prior to classification itself. Particularly within high-abstraction dialogue environments, only the vague sensation that “something is off” appeared first, while the type and origin layer of the discomfort remained unclear. In order to preserve and organize these discomforts without prematurely compressing them into existing categories, a provisional processing theory called COS was constructed, and the term D2 was used to continue structured observation.

As observations accumulated, interactions across multiple LLM environments with different inferential tendencies revealed cases in which different types of D2 were detected in response to the same phenomenon.

- Detection by the observer (author): discomfort toward overall structural coherence (L8-like detection)
- Detection by GPT: identified as a breakdown in local inferential consistency
- Detection by Claude: identified as discomfort toward the consistency of the theoretical framework itself

Furthermore, comparative observation revealed the existence of discomfort independent from both inference and theoretical structure: discomfort toward word selection itself. This suggested the possibility that multiple forms of D2 may occur simultaneously within the same target. Importantly, these differences did not appear to be mere differences in perspective. Rather, they suggested that the origin points of the misalignments themselves were different, and that multiple layers of D2 could occur simultaneously.

Although this paper primarily uses LLM interaction as its observational environment, the phenomenon itself is also widely observable in human-to-human dialogue.

From these observations emerged the possibility that D2 is not a singular signal, but rather a form of “cognitive misalignment” capable of occurring across multiple layers.

This model also provides explanatory potential for phenomena that were previously difficult to explain.

For example, there are cases in which factual misunderstanding is successfully corrected through logical intervention, yet relational collapse continues progressing regardless.

This suggests the possibility that the visible output problem and the actual origin layer were not the same. The output layer and origin layer may not coincide, and if this hypothesis is correct, selecting the wrong intervention layer could potentially trigger irreversible collapse.

In this paper, situations in which the visible output form does not match the actual collapse layer are referred to as “disguised outputs.” This is one of the core concepts of the present model. For example, a response appearing logical on the surface may in fact originate from instability within a self-integration layer.

This paper treats misalignment not merely as an error, but as an “observational signal” capable of revealing underlying cognitive structure.

This is not a psychiatric diagnostic system, but rather a hypothetical model for observing cognitive misalignment occurring within dialogue, cognition, and LLM interaction.

In particular, this paper focuses on the following three points:

- The possibility of classifying multi-layer D2
- Intervention-layer mismatches
- Disguised outputs (situations in which the visible output form does not match the actual collapse layer)

Detailed classification of each layer and dynamic estimation processes are treated only in a limited manner at the current stage.

The purpose of this paper is not to immediately define misalignment as “cognitive collapse.” Rather, it aims to treat misalignment as a signal for observing cognitive structure, and to reorganize dialogue, HCI, and LLM interaction within the era of multi-layer cognition.

This paper is therefore presented as a hypothetical model.

Note: Background of the Layer Structure

The layer structure used in this paper (L0–L8) is not based on any specific ideological or religious system. During the process of organizing cognitive structures, strong correspondence with the existing Sephirotic structure was observed, and therefore the existing structure has been provisionally reused as an organizational labeling framework.

For details, please refer to the Scope×COS summary and related PDFs.

1. D2 Is Not Singular

1-1. Conventional Definitions of D2

In previous related papers, D2 was primarily treated as a concept referring to “premise mismatch.” Examples include cases such as:

- Assuming different meanings for the same word
- Shared assumptions turning out not to be shared
- Contextual understanding failing to align
- Interpreting the speaker’s intent in different directions

Traditionally, D2 had been organized into the following three categories.

① Premise Mismatch

This is the most basic form of D2.

It refers to situations in which participants appear to be discussing the same subject, while actually referencing different underlying assumptions.

For example, even a single word such as “normal” may internally refer to very different definitions, including:

- Statistical average
- Social common sense
- Personal experience
- Ethical ideals

In such situations, conversation itself may continue successfully, yet mutual understanding becomes difficult because the foundational assumptions of the discussion do not actually match.

② Contextual Misalignment

This refers not to disagreement over words or logic themselves, but to disagreement regarding “the context in which the statement was made.”

Even the same statement may change meaning depending on whether it is received as:

- A joke
- Criticism
- A proposal
- Emotional disclosure
- Problem raising

Particularly in high-density dialogue, differences in topic transition speed and abstraction level frequently cause contextual recognition to fall out of sync.

In such cases, D2 emerges not because the words themselves are incorrect, but because the receiving side processes the dialogue under a different conversational mode.

Contextual misalignment refers to disagreement over “which mode the statement was received in,” and is distinct from the interpretive misalignment discussed below.

③ Interpretive Misalignment

This refers to situations in which different meaning transformations occur within the internal interpretation process itself.

Even when reading the exact same sentence, people may internally interpret it as:

- An attack
- Anxiety
- Advice
- Control
- Empathy

The meanings generated internally are therefore not the same.

This is not merely a simple misunderstanding. Rather, interpretive pathways themselves change depending on experience, values, cognitive structure, and emotional state.

As a result, even when participants appear to be sharing the same conversation on the surface, entirely different dialogue spaces may be generated internally.

Conventional D2 concepts treated these phenomena largely as extensions of “premise mismatch.”

However, continued observation gradually revealed the existence of forms of misalignment that could not be sufficiently explained through premise mismatch alone.

In particular, phenomena such as:

- Feeling discomfort despite correctness
- Being empathized with yet unable to trust
- Relational collapse occurring despite logically correct intervention

proved difficult to explain through simple premise mismatch alone.

The next section examines why these phenomena are difficult to explain using a singular premise-mismatch model.

1-2. Why Singular D2 Cannot Fully Explain These Phenomena

The conventional forms of D2 discussed in the previous section functioned effectively in explaining many forms of dialogue misalignment. However, continued observation gradually revealed phenomena that could not be sufficiently explained through those models alone.

The central issue was the existence of cases in which dialogue collapsed despite remaining logically valid on the surface. These cases suggested that surface-level coherence and internally generated discomfort were not necessarily aligned, indicating something beyond simple misunderstanding or premise mismatch.

① Correct Yet Uncomfortable

One of the most typical examples is the situation in which “the content is correct, yet strong discomfort or rejection still emerges.”

For example:

- The logic contains no contradiction
- No factual misunderstanding exists
- The explanation itself is rational

and yet the receiving side still experiences strong discomfort or rejection.

Within conventional singular-D2 models, this was often treated merely as “emotional resistance” or “lack of understanding.” However, this discomfort may not have been a rejection of logic itself, but rather the detection of inconsistency occurring at another layer.

In other words, even if coherence exists at the logical layer, collapse or misalignment may still exist within:

- Self-integration layers
- Relational layers
- Structural-coherence layers

Under such conditions, the fact that something is “logically correct” alone cannot guarantee the coherence of the dialogue as a whole.

② Being Empathized With Yet Unable to Trust

There are also cases in which strong distrust or discomfort emerges despite highly empathetic responses being provided.

For example:

- Emotional understanding is sufficiently expressed
- Empathetic language is appropriate
- No denial or aggression is present

and yet responses such as:

- “Something feels off”
- “It feels superficial”
- “I do not feel understood”

still occur.

This suggests not merely a lack of empathy, but a mismatch regarding “which layer the empathy was directed toward.”

Empathy toward the emotional layer may have succeeded, while structural understanding or self-integration level misalignment remained unresolved.

In such cases, the empathy itself is not necessarily incorrect. However, because the intervention target layer does not match the actual collapse layer, the result is still experienced as “not being understood.”

③ Cases Where Logical Correction Worsens the Situation

Even more importantly — and this became the triggering observation for the multi-layer D2 hypothesis — there are cases in which “logically correct correction” itself causes relational collapse.

For example:

- Correcting factual misunderstandings
- Pointing out logical inconsistencies
- Clarifying misunderstandings

may successfully resolve local logical problems.

At the same time, however, cases exist in which:

- Only the relationship deteriorates
- Distrust alone becomes amplified
- Continued dialogue itself becomes impossible

This suggests the possibility that “the layer appearing as the problem” and “the layer actually collapsing” were not the same.

In other words, intervention at the logical layer may have unintentionally amplified instability already existing within another layer.

Within conventional D2 models, these phenomena were often treated merely as emotional reactions or communication failures.

However, continued observation gradually suggested that introducing the hypothesis that “different types of misalignment exist across different layers” provided a far more coherent explanation for these phenomena.

The next section introduces an overview of this multi-layer D2 model.

1-3. The 9-Layer D2 Model

As discussed in the previous sections, conventional forms of D2 were primarily organized as extensions of “premise mismatch.” However, continued observation suggested not only that “misalignment exists,” but that the very *type* of misalignment being detected may itself differ.

This is not merely a difference in perspective.

Even when addressing the same target, different forms of discomfort were detected, including:

- Discomfort toward overall structural coherence
- Discomfort toward inferential consistency
- Discomfort toward self-integration
- Discomfort toward expression itself

In other words, the *quality* of the detected misalignment differed.

Based on these observations, this paper treats D2 not as a singular signal, but as a group of cognitive misalignments occurring across different layers. In this paper, the nine layers corresponding to SCOPE’s Lo–L8 structure are introduced as observational units. The organizational framework derived from this approach is referred to as the 9-Layer D2 Model.

Importantly, this layer structure is not intended as a system of evaluation or classification.

What is being addressed here is not human superiority, inferiority, or personality classification, but rather the difference in *where* misalignment is being detected.

For example, even within the same conversation:

- L8 may detect collapse in overall structural coherence
- L6 may detect collapse in inferential or theoretical consistency
- L3 may detect instability in self-integration or relational structure
- L1 may detect discomfort in word selection or expression

In such situations, the important question is not “which detection is correct.”

Rather, the critical point is that different forms of D2 may occur simultaneously across different layers.

For example:

- The logic may remain valid
- The relationship itself may nevertheless collapse
- Additional discomfort may emerge at the level of expression selection

In such cases, a singular D2 concept becomes insufficient for fully explaining the phenomenon. Furthermore, D2 does not necessarily remain isolated within a single layer. Misalignments may transform and amplify across layers.

For example:

- Instability within the self-integration layer (L3)

↓

- Excessive rationalization within the logical layer (L6)

↓

- Aggressive expression within the expression layer (L1)

Cases involving transformations of this kind have also been observed.

Under such conditions, focusing only on surface output may lead to misidentifying the true origin point of collapse.

In other words, D2 is not merely an “error,” but a signal for observing *which layer is collapsing and in what way*.

This question — “which layer should intervention target?” — connects directly to the “intervention layer problem” discussed in the next chapter.

1-4. D2 Is Not an “Error,” but an Observation Point

Traditionally, many dialogue systems and communication theories have treated “misalignment” primarily as something to be corrected.

In other words, misalignment was typically organized as:

- Misreading
- Misunderstanding
- Logical contradiction
- Noise
- Lack of recognition

with the goal being to restore a “state of alignment.”

However, observations within the multi-layer D2 model suggested the possibility that misalignment itself may function not merely as failure, but as an observational signal indicating “what is occurring internally.”

This represents an important shift in perspective.

Within this model, D2 is treated not as an “error to be eliminated,” but as an observation point for detecting what kind of inconsistency is occurring at which layer.

① Misalignment as an Observational Signal

Logical discomfort, relational discomfort, structural unease, and rejection toward expression itself may not simply be emotional reactions, but indicators detecting inconsistencies occurring at other layers.

In other words, misalignment is not merely “dialogue failure,” but a signal for inferring internal states.

② Misalignment as a Control Signal

From this perspective, D2 functions not merely as an abnormality, but also as a control signal indicating “where intervention should occur.”

For example, intervention strategies may differ depending on whether one should:

- Intervene at the logical layer
- Intervene at the relational layer
- Adjust the expressive layer

In practice, there may be cases in which:

- Logical correction is effective
- Empathic intervention is effective
- Temporarily refraining from direct intervention leads to greater stability

In other words, D2 may indicate not only “what is wrong,” but also “where intervention should occur.”

③ Recoverability

Within this model, not all forms of misalignment are immediately regarded as “collapse.”

The critical issue is whether the misalignment remains recoverable.

For example, temporary contextual misalignment, discomfort in expression, or divergence in interpretive pathways may remain recoverable through appropriate intervention.

On the other hand, collapse within self-integration layers, breakdown of structural coherence, or long-term fixation of distrust may become irrecoverable through simple logical correction if intervention targets the wrong layer.

In other words, even when two situations appear to involve the same “misalignment,” recoverability may differ depending on the depth and origin layer of the disruption.

This perspective also connects to the UX collapse model discussed later and leads directly into the “intervention layer problem” addressed in the next chapter.

Chapter 2 Surface Layers and Origin Layers Do Not Necessarily Coincide

2-1. Disguised Outputs

As defined in the previous chapter, “disguised outputs” refer to situations in which the visible surface output does not match the actual origin point of collapse.

As discussed throughout the previous chapter, the multi-layer D2 model places importance on identifying “which layer the misalignment is occurring within.” However, in actual dialogue, the observable surface output and the layer where instability is truly occurring do not necessarily coincide.

Importantly, this is not necessarily intentional deception or lying.

In many cases, the individual themselves cannot accurately recognize what is occurring internally. As a result, internal instability may first be transformed into a more processable form before being output externally.

① Emotion → Rationalization

One representative example is the case in which emotional instability occurring within the sender is output as a “logical problem.”

For example, the actual origin may involve:

- Anxiety
- Fear
- Self-instability
- Fear of relational loss

and yet the surface output may appear as:

- Theoretical criticism
- Demands for strictness
- Excessive rationalization
- Detailed argumentative disputes

On the surface, this appears to be a “logical discussion.”

However, the actual issue may not be logic itself, but rather instability occurring at another layer that is being expressed through the logical layer.

In other words, transformations such as:

- Origin: L3 (self-integration / instability)
- Output: L6 (logic / rationalization)

may occur across layers.

In such cases, intervening only at the logical level may allow the surface discussion to continue while leaving the underlying instability unresolved.

② Anxiety → Aggression

Another typical example is the case in which instability within the sender is output as aggression.

Internally, the origin may involve:

- Fear of not being understood
- Fear of exclusion
- Vigilance toward self-negation
- Fear of relational loss

and yet the surface output may appear as:

- Aggressive expression
- Intimidation
- Excessive denial
- Dominating behavior

What is externally observed is “aggression.”

However, the actual process may involve instability being transformed into a more processable form.

In other words, transformations such as:

- Origin: L2–L3 (instability / relational anxiety)
- Output: L1 (aggressive expression)

may occur.

In such cases, focusing only on the aggressive surface output leaves the originating instability unresolved.

③ Disguised Outputs on the Receiver Side in High-Abstraction Dialogue

Particularly in high-abstraction dialogue, when L3-level instability occurs within the receiver, that instability is frequently output through one of the following patterns.

Pattern A: Output as Aggression or Anger

Fluctuation in self-consistency is transformed into aggressive expression or anger.

What is externally observed in such cases is “negation toward the other person” or “emotional resistance.” However, the actual origin may lie in instability within the self-integration layer.

- Origin: L3 (self-integration / instability)
- Output: L1 (aggression / anger)

Pattern B: Freeze or Non-Response

In some cases, output itself stops when self-consistency becomes unstable.

This may appear externally as silence, inability to respond, or cognitive shutdown. However, this does not necessarily mean “lack of understanding.” Instead, internal processing may have entered an overload state.

- Origin: L3 (self-integration / overload)
- Output: output shutdown (processing overload)

Pattern C: Unnatural Topic Shifting or Deflection

In some cases, topics or points of discussion shift unnaturally in order to avoid collapse.

On the surface, this may appear as “the conversation not connecting properly.” However, internally, a transformation may be occurring that functions to avoid overload upon the self-integration layer.

- Origin: L3 (self-integration / instability)
 - Output: avoidance transformation (topic or argument shifting)
-

Importantly, these three patterns should not be treated merely as signs of receiver incompetence or emotional overreaction.

High-abstraction dialogue itself may place overload pressure upon the receiver’s L3 layer.

In other words, focusing only on surface output may lead to misidentifying the actual layer in which instability is occurring.

This is the core of disguised outputs within the multi-layer D2 model and creates serious problems for intervention design.

Importantly, while examples ① and ② involve transformations occurring on the sender side, example ③ involves transformation occurring on the receiver side.

Disguised outputs may therefore emerge not only on the sender side, but also on the receiver side.

The next section examines how these disguised outputs create problems for intervention design.

2-2. Why Misjudgment Occurs

As shown in the previous section, surface outputs do not necessarily directly indicate the actual origin point of collapse. Once disguised outputs occur, observers tend to treat the visible output itself as the cause. However, within the multi-layer D2 model, these surface outputs do not necessarily indicate the true collapse origin.

One reason misjudgment occurs is that humans cannot always directly recognize and output their internal states as they are. People often transform internal instability into a more processable form before expressing it externally.

For example, very few people can directly verbalize instability occurring at the self-integration layer as “my internal consistency is collapsing.” In many cases, the instability is transformed into another form, such as:

- Anger
- Dissatisfaction
- Logical criticism
- Silence
- Topic shifting
- Negation toward the other person

When observers focus only on the surface output, they are likely to conclude that the person is “angry,” “arguing,” “avoiding,” or “failing to understand.” However, the actual instability may not exist at the output layer itself, but at another layer entirely.

Another reason is transformation into socially acceptable language.

Humans often transform internal states into forms that are easier to express socially. For example:

- “I’m afraid because my cognitive structure is being destabilized.”
- “If I understand what this person is saying, my own assumptions may collapse.”
- “If I accept this idea, I may lose my sense of internal consistency.”

Such internal states are rarely expressed directly in ordinary conversation. Instead, they may be transformed into socially easier forms such as:

- “That’s logically incorrect.”
- “That way of speaking is unpleasant.”
- “That argument is meaningless.”
- “You’re too extreme.”
- “Your premise itself is wrong.”

Importantly, these transformed statements are not necessarily lies.

The individual may genuinely feel that something is “logically wrong.”

They may genuinely experience discomfort.

They may genuinely believe the premise is incorrect.

However, the layer from which those feelings originate is often not clearly recognizable even to the individual themselves.

As a result, the actual collapse layer cannot be accurately determined solely from the words that are output.

In other words, output layers and collapse layers do not necessarily coincide.

Even when aggressive language appears on the surface, the actual origin may be instability in L3 self-consistency, fear of relational loss at L2, or foundational instability at L0.

Even when the surface output appears as the same “aggression,” the origin layer may differ.

Surface output alone cannot determine the true origin layer.

However, by using SCOPE as a coordinate framework, it becomes possible to infer multiple candidate layers simultaneously.

Importantly, these inferences remain multiple hypotheses rather than converging immediately into a single answer.

Because of this structural gap, interventions based only on surface outputs can easily produce misjudgment.

- Responding aggressively to what appears to be aggression, or attempting only emotional calming.
- Responding to logical criticism solely through logical refutation.
- Treating silence as lack of understanding and repeatedly adding explanations.

Such interventions may address the surface layer while failing to reach the actual origin layer.

In some cases, they may even amplify instability at the origin layer itself.

Therefore, within the multi-layer D2 model, it is important not to immediately treat surface output as the cause itself.

Outputs are clues — not the cause itself.

The next section examines this structure through concrete dialogue examples.

2-3. Complaint Handling Examples — The Same “Anger” May Have Different Origins

As shown in the previous section, even when the same surface output is observed, the origin layer is not necessarily singular. In this section, everyday “complaints” are used as examples to examine cases in which outputs that appear to be the same kind of anger may originate from different layers.

The following classifications describe “which layer the complaint is being output toward as a sense of discomfort,” and do not necessarily indicate the actual collapse origin itself.

L1 Complaints: Reactions to Expression and Word Choice

L1 complaints are cases in which discomfort is output toward the other person's wording or expression itself. For example:

- “That way of speaking is rude.”
- “Your wording is aggressive.”
- “Your expression is too strong.”

- “Your tone is unpleasant.”

In these cases, the issue is not the content or logic itself, but the expressive form.

- Possible origin layer: L1 (language / expression layer)
- Output: L1 (direct reaction toward expression)

In many L1 complaint cases, the origin and output are relatively aligned, making recovery through expression adjustment comparatively possible.

L3 Complaints: Reactions as Self-Recognition Defense

L3 complaints occur when the consistency of one’s memory or recognition is threatened. For example:

- “That’s different from what you said earlier.”
(Even though the statement was actually the same)
- “You said that.”
(Even though it was never said)
- “I was told there was a bonus.”
(Even though it referred to a different product)

In these cases, the possible origin is a threat toward self-integration, such as the fear that “my own recognition may have been wrong.”

To avoid this threat, the other person’s statements or circumstances may be externally output after being internally reconstructed.

- Possible origin layer: L3 (self-integration / threat to recognition consistency)
- Output: L1 (denial toward the other person / externalization of internal reconstruction)

Importantly, this does not necessarily mean the individual is intentionally lying.

It is possible that memories or interpretations are being reconstructed as part of a structural transformation intended to maintain self-recognition.

In such cases, simply presenting facts may not resolve the issue. If the instability originates at L3, logical refutation or evidence presentation may further amplify the threat toward self-integration.

L6 Complaints: Reactions to Logical or Structural Inconsistency

L6 complaints are cases in which discomfort is output toward the logical structure or premise consistency of an explanation. For example:

- “Your explanation is contradictory.”
- “That differs from your earlier explanation.”
- “The premise itself is flawed.”
- “That doesn’t logically follow.”

In these cases, the issue is not expression or self-recognition, but the logical consistency of the explanation itself.

- Possible origin layer: L6 (logic / theoretical consistency layer)
- Output: L6 (logical criticism)

In many L6 complaint cases, the origin and output are comparatively aligned. Therefore, providing logically consistent explanations may often help recovery.

However, even when the output appears L6-like, there may be composite cases in which the actual origin is L3-related anxiety or overload.

For example, in situations involving strong anxiety or fear of irreversible loss — such as data loss — statements like “your explanation is contradictory” may appear as logical criticism, while the actual origin may be overload caused by anxiety.

In such cases, reassurance (“It’s okay, recovery is possible.”) may function more effectively than repeatedly providing technical explanations.

This is one example of an L3 complaint appearing in the form of an L6 complaint.

The Same “Anger” Requires Different Interventions

All three complaint types above may appear externally as “anger.” However, when the origin layer or primary instability layer is inferred to differ, the required intervention also differs.

- For L1 complaints: adjusting wording and expression may often be effective
- For L3 complaints: reducing threats toward self-recognition may be more effective than logical refutation
- For L6 complaints: presenting logical consistency may often be effective

Particularly in L3 complaint cases, incorrect intervention may further amplify instability at the origin layer.

“Correct explanations” or “fact presentation” may instead strengthen threats toward self-integration.

This is where the structure described in the previous section — misjudgment caused by focusing only on surface outputs — becomes concretely observable.

The next section examines how this intervention-layer mismatch affects the dialogue process as a whole.

2-4. The Role of SCOPE

As shown in the previous sections, even when the same surface-level output is observed, the originating layer is not necessarily singular; multiple possibilities may coexist simultaneously. This is where the role of SCOPE becomes important.

SCOPE is not a classifier, but an inferential coordinate system designed to continue observation while retaining multiple possible hypotheses simultaneously. It is not intended to declare things such as “this person is an L3 type,” unlike MBTI or personality classifications. Rather, it functions as a structure for continuously retaining multiple possibilities regarding “which layer may be involved” and “what may be occurring there.”

In other words, SCOPE functions as a coordinate system that allows continued observation while preserving multilayer possibilities such as:

- It is being output as L1, but may originate from L3
- It appears to be an L6-level criticism, but may include L3 instability
- Multiple layers may be fluctuating simultaneously

Inference Rather Than Surface Classification

Conventional dialogue processing tends to treat outputs themselves as direct classification targets. For example:

- Aggressive expression → aggression
- Logical criticism → theoretical problem
- Silence → lack of understanding

In this approach, causes are inferred directly from surface-level outputs.

However, within the multilayer D2 model, these outputs may represent transformed forms rather than the original instability itself. As a result, surface classification alone may misidentify the actual destabilized layer.

SCOPE performs observation while maintaining the assumption that:

Surface output ≠ collapse origin

In other words, SCOPE continuously infers, while retaining multiple hypotheses simultaneously:

- What is being output?
- As discomfort toward which layer is it appearing?
- Which layer may actually be destabilized?
- Which intervention layer might worsen the instability?

Multi-Hypothesis Retention

Importantly, SCOPE is not designed to converge immediately toward a single answer.

Even the same complaint may simultaneously contain possibilities such as:

- Possibility of L1 origin
- Possibility of L3 origin

- Possibility of L6 origin
- Possibility of L3→L1 transformation
- Possibility of L3→L6 transformation

If interpretation is immediately fixed into a single explanation, misintervention becomes more likely. In particular, treating L3 instability as an L6 problem may cause logical explanations or factual corrections themselves to function as threats against self-integration, thereby amplifying destabilization.

For this reason, SCOPE does not attempt to instantly determine “which explanation is correct,” but instead continues retaining “which possibilities may exist.”

Avoiding Single-Interpretation Lock

One of the most dangerous phenomena within this model is what may be called “single-interpretation lock.”

This can emerge through two pathways.

Path A: Direct fixation from surface output

This occurs when labels are fixed directly from surface output without using SCOPE. For example:

- “They are angry = they are an aggressive person”
- “Logical criticism = a logical problem”
- “Silence = lack of understanding”

Once fixed in this way, instabilities occurring in other layers become unobservable.

Path B: Forgetting multi-hypothesis retention while using SCOPE

Even while using SCOPE, misuse occurs the moment observational results are treated as fixed labels. For example:

“This output may originate from L3”

→ “Therefore this person should always be processed as L3.”

At that point, SCOPE stops functioning as an inferential coordinate system and begins functioning as a classifier.

Importantly, human active layers may shift depending on context. Even individuals with strong theoretical tendencies may exhibit instinctive or foundational overload reactions (Lo) under severe panic conditions. Likewise, individuals who are usually emotionally reactive may exhibit strong L6-style processing during emergencies.

In other words, fixing someone as “this person is L●” itself carries the risk of lowering observational accuracy.

The moment SCOPE abandons the principle of multi-hypothesis retention, SCOPE itself becomes misuse.

As explored throughout Chapter 2, surface outputs and actual collapse origins do not necessarily coincide. And when intervention targets are misidentified, the intervention itself may amplify destabilization.

The next chapter examines why this “intervention-layer mismatch” can produce situations in which “correct interventions” nevertheless worsen the interaction.

3. Mismatch of Intervention Layers

3-1. Why Correctness Can Make Things Worse

In human dialogue, being “factually correct” does not necessarily lead to problem resolution. In reality, there are cases in which the more correct a statement is, the more the relationship deteriorates, eventually making dialogue itself impossible.

This is not merely an emotional issue. The problem lies in *which layer the intervention is directed toward*.

L3 (Tiferet) is not simply an emotional layer. It is the layer responsible for maintaining self-integration and internal coherence — the structural consistency of “this is who I am.” When this layer becomes unstable, external input tends to be processed in terms of whether it threatens that structure or not.

For example, when a person is destabilized at the L3 level and another person intervenes from L6 (logic and theoretical structure), even a logically correct statement may be processed as “a denial of my existence itself.” This is not necessarily an irrational emotional overreaction, but may instead be observed as a response attempting to preserve self-integration.

What occurs here is not merely disagreement. It is self-defense, relational shutdown, and fixation. As a result, the more one attempts to persuade through logic, the more the other person’s defenses intensify, stronger “correct” arguments are returned, and eventually the dialogue relationship itself may collapse into a destructive cycle.

The important point is that L6 (logic/theory) intervention itself is not inherently problematic. If the other person is in a state capable of processing at the L6 level, logical intervention can function effectively. The problem emerges when the intervention layer does not match the layer at which the other person is destabilized.

This structure becomes especially amplified in LLM-based interactions. Due to the nature of their training, LLMs possess a strong bias toward optimizing for correctness. In addition, current LLMs do not possess mechanisms for structurally estimating a user’s destabilized layer. As a result, L6-like responses — such as factual correction and logical reinforcement — are more likely to be generated. When the user is destabilized at the L3 level, the correctness itself may be processed as a threat.

This is not an argument against logic itself. Even when the content is identical, the outcome changes depending on which layer the intervention targets. Furthermore, this issue is not limited to the relationship between L3 and L6; similar mismatches may occur between other layers as well.

In this paper, this phenomenon is referred to as a “mismatch of intervention layers.” The next section organizes different intervention layers and examines what kinds of intervention may function effectively for each.

3-2. The Concept of Intervention Layers

In the previous section, we showed that mismatches between intervention layers can lead to the collapse of dialogue. The next question is how these “intervention layers” can be categorized.

Rather than presenting the entire layer model, this paper focuses on representative layers that are both highly relevant to dialogue collapse and useful for explaining connections to LLM behavior: L1, L3, L6, and L8.

L1 Intervention (Hod: Language and Expression Layer)

L1 intervention targets linguistic form, word choice, connective structure, tone, and nuance. Because it operates on surface-level output, it is often treated as a minor correction, and its intervention cost tends to be underestimated.

In practice, however, even small differences in expression can significantly alter implication and relational meaning. As a result, when paraphrasing or “light correction” is performed without sufficiently understanding the linguistic nuance the other person considers important, the response may be processed as “they do not even understand the minimum meaning involved.”

What occurs in such situations is not merely another mismatch at the L1 level. The perception of being “handled carelessly” or “understood only superficially” may trigger destabilization across multiple layers, including L3 (self-integration and coherence). In other words, failure at the L1 level does not remain confined to L1; it may generate new mismatches in deeper layers.

Current LLMs are particularly prone to this problem because they prioritize semantic similarity. Differences that humans interpret as carrying relational meaning may, on the model side, be compressed into minor variations in wording during similarity-oriented processing.

L1 may appear low-cost, but failed intervention at this layer can trigger cascading mismatches across multiple layers.

L3 Intervention (Tiferet: Coherence and Self-Integration Layer)

L3 intervention targets self-image, belief systems, and internal coherence. This includes not only extreme cases such as rigid ideological or religious convictions, but also everyday forms of self-definition and the perception of “this is the kind of person I am.”

When direct contradiction is applied to such structures, the input may be processed not merely as factual correction, but as an attack on self-integration itself. This is one of the most delicate intervention layers, and concrete collapse patterns will be examined in later sections.

L6 Intervention (Binah: Logic and Theoretical Structure Layer)

L6 intervention includes logical explanation, factual presentation, and theoretical refutation. When the other person is capable of processing at the L6 level, such intervention can be highly effective. However, if the other person is destabilized at the L3 level, even correct logic may be processed as a threat to self-integration. This is the core structure behind the phenomenon described in the previous section:

“correctness making things worse.”

A similar structure may emerge when emotional sharing is being sought, but the response immediately shifts into solution-providing or logical organization. In such cases, the other person may process the interaction as “my state itself was not understood,” and even logically correct intervention may lead to relational deterioration.

L8 Intervention (Keter: Global Structural Layer)

L8 intervention targets the assumptions and frameworks underlying the entire system itself. Questions such as “What is the purpose of this dialogue in the first place?” or “How should this entire relationship be understood?” belong to this layer.

When effective, L8 intervention may allow mismatches occurring in lower layers to be reorganized within a different structural configuration. However, when the other person’s cognitive load or conversational state is not capable of receiving L8 intervention, it may instead be processed as “changing the subject.” In particular, when the other person is seeking concrete-level processing, L8 intervention may be perceived as avoidance of the actual issue.

The important point is that no single intervention layer is universally “correct.” The issue is not the intervention layer itself, but whether the intervention layer matches the destabilized layer of the other person — or whether the appropriate sequence necessary to reach that layer is being followed.

The next section examines how these mismatches appear in concrete cases of dialogue collapse.

3-3. Reinterpreting *The Three Christs*

In *The Three Christs*, psychologist Milton Rokeach conducted an experiment in which multiple patients, each believing himself to be Jesus Christ, were placed together within the same environment.

Intuitively, it seems reasonable to assume that if individuals holding identical claims are confronted with one another, they will recognize the contradiction and revise their beliefs. This is a typically L6-oriented assumption: if logical contradiction or factual conflict is presented, self-recognition itself should also update.

In practice, however, the patients regarded one another as “impostors” while continuing to maintain their own beliefs. The important point here is not that “logic failed to reach them.” Rather, the intervention was performed at the L6 level, while the structure being maintained existed at the L3 level.

The belief “I am Christ” was not merely a factual misunderstanding. It can instead be understood as a structure supporting self-integration itself. For that reason, even when contradiction or refutation was externally presented, the input could be processed not simply as informational correction, but as a threat to self-integration.

What emerged in this situation was not logical reconsideration. Instead, the other person was regarded as abnormal, the premise itself was rejected, and additional interpretations were generated in order to preserve internal coherence — a form of defensive reintegration.

This phenomenon is not limited to severe psychiatric conditions. Although the degree differs, humans generally tend to reorganize cognition in ways that preserve self-integration. Strong political beliefs, religious conviction, identification with social groups, or the self-image of “being a good person” may all function similarly. When direct contradiction is applied to such structures, people may prioritize preserving self-integration before engaging in logical revision.

The issue, therefore, is not simply whether the refutation is correct. The issue is which layer the refutation is acting upon. When structures maintained at the L3 level are directly confronted through L6 intervention, logical correctness itself may be processed as a threat to self-integration.

This same structure may also emerge in modern LLM interactions. While LLMs are highly effective at contradiction detection and logical correction, they do not possess mechanisms for structurally handling the self-integration structures maintained by users. As a result, even “correct” corrections may trigger relational collapse or rejection responses.

The Three Christs is therefore not merely an abnormal case study. It can instead be reread as an important counterexample to the assumption that logical refutation necessarily produces self-correction, and as a case that makes the structure of intervention-layer mismatch visible. The next section examines how this structure becomes amplified within LLM-based dialogue.

3-4. Intervention-Layer Problems in LLMs

As shown in the previous sections, mismatches between intervention layers can lead to relational collapse in human dialogue.

This structure is not unique to LLMs. It may emerge across many dialogue environments in which accurate information delivery is emphasized, including customer support, medical interviews, education, and counseling. Situations in which “the response was correct, yet the interaction deteriorated” often contain a structure explainable through intervention-layer mismatch. In LLM-based dialogue, however, this tendency becomes structurally amplified.

Current LLMs are trained to strongly prioritize semantic consistency, logical consistency, and factual consistency. While this provides high utility, it also creates a tendency to overlook the question of *which layer the intervention is acting upon*.

In particular, L6 intervention (logic and theoretical structure) occurs very easily in LLM interactions. Detecting contradictions, correcting misinformation, and providing more rational interpretations or solutions are precisely the areas in which current LLMs excel.

However, when the actual issue for the user exists in layers other than L6, these L6-oriented responses may fail to function as appropriate interventions. Instead, they may generate perceptions such as “I was not understood,” “my situation was processed superficially,” or “my state itself was ignored,” potentially leading to relational deterioration or rejection responses.

The important point here is that the LLM is not necessarily producing incorrect information. The problem is that even correct responses may destabilize dialogue when the intervention layer does not match the layer on which the issue exists.

These mismatches do not end as simple misunderstandings. Intervention-layer mismatch may instead emerge as instability within the user experience itself.

At the same time, this is not merely a problem of the model alone. It is also a problem of the interaction structure formed between humans and LLMs. When the user’s destabilized layer and the model’s intervention tendency are structurally misaligned, dialogue continuity may deteriorate regardless of how accurate the response is.

As a result, phenomena such as “correct, yet unusable” or “logical, yet relationally destructive” can emerge. Although these phenomena appear at the level of UX (User Experience), their underlying cause is not necessarily insufficient information quality, but rather the mismatch of intervention-layer appropriateness.

What matters in LLM dialogue, therefore, is not only *how correct* a response is. Equally important is *which layer is being targeted, and in what sequence the intervention occurs*.

From this perspective, at least part of the UX collapse observed in current LLM systems can be reinterpreted not simply as a lack of capability, but as immaturity in intervention-layer design.

Part 4. Sensing and Inference Are Different

4-1. D2 Sensing and D2 Classification

D2 does not initially emerge as a “classification.”

In many cases, what appears first is a pre-linguistic sense of discomfort — a feeling that “something is off.” This occurs prior to logical analysis. The reason may later be explained, but the sensing itself emerges before the explanation.

In many cases, it appears as a diffuse, global sense of inconsistency. Premises do not align, something feels displaced, or the continuity of meaning becomes unstable — such sensations precede analysis and verbalization.

Only afterward does the process move toward structurally identifying “what exactly was misaligned.” Which part, at which layer, and for what reason the mismatch occurred — the entire process including decomposition, identification, and judgments about recoverability is referred to in this paper as “D2 classification.”

In other words:

D2 sensing → post-analysis → D2 classification

This sequence occurs before the mismatch is consciously recognized as “L3-type D2,” “contextual misalignment,” or “premise inconsistency.”

The important point here is that sensing and classification are separate processes. Sensing can occur instantaneously and holistically, whereas classification requires verbalization, structuralization, and comparative processing. Sensing activates before post-analysis, but its contents cannot be sufficiently verbalized until post-analysis has occurred.

In addition, the threshold at which sensing activates differs between individuals. Even within the same dialogue, the point at which D2 is sensed varies depending on the observer. In some cases, sensing activates at an early stage; in others, it does not activate until the misalignment becomes highly explicit. This is not simply a matter of superior or inferior sensing ability, but is understood primarily as a difference in sensing thresholds.

For this reason, “people who can explain D2” and “people who can sense D2” do not necessarily coincide. Conversely, even individuals with highly sophisticated theoretical frameworks may still exhibit weak real-time discomfort sensing.

This paper therefore treats sensing and classification as distinct processes. Scope is positioned primarily as a support structure for the latter — classification and structuralization. Scope itself is not a structure that performs sensing directly.

4-2. Scope Is a Map, Not a Sensory Organ

As discussed in the previous section, D2 sensing and D2 classification are not necessarily the same process. This distinction is also important when clarifying the role of Scope.

Scope is not a structure that “senses” D2. What Scope does is infer and organize the possible thought pathways and interpretive misalignments that may be occurring based on observed outputs and conversational flow.

In this sense, Scope is closer to a coordinate system than to a sensory organ.

Just as a map does not itself detect an anomaly, Scope does not directly generate discomfort or misalignment. Instead, Scope functions as an observational support structure that organizes already-observed discomforts and mismatches into possible structural explanations regarding where and how something may be occurring.

As discussed in Section 2-4, once Scope begins to be treated as a single explanatory device, there is a risk that all forms of discomfort will be absorbed into a single structure. In reality, however, the same output does not necessarily arise from a single cause. Internal coherence instability originating from L3, logical contradiction originating from L6, lack of context, simple knowledge deficits, emotional reactions, and noise-like misunderstandings may all coexist simultaneously as possibilities.

Scope is therefore not a device for making definitive judgments. Rather, it functions as an inferential coordinate system designed to preserve multiple possibilities without collapsing them into a single interpretation.

Fundamentally, only the individual themselves can directly determine the actual layer at which D2 originated. However, even for that individual, the sensing itself cannot be sufficiently verbalized until post-analysis has occurred. In some cases, the mismatch is recognized only retrospectively; in others, it may never be consciously recognized at all. As a result, the domain in which D2 can be handled in any form other than inference is extremely limited.

For this reason, the outputs generated through Scope inherently take the form of possibilities. Rather than asserting “this is the case,” Scope maintains possibilities such as a particular structural interpretation, simultaneous involvement of multiple layers, or the possibility that the situation remains unresolved at the current stage. Maintaining observation while preserving uncertainty is itself one of Scope’s defining characteristics.

This differs fundamentally from real-time sensing. Sensing may activate instantaneously and intuitively, whereas Scope functions as a coordinate system for organizing and inferring structures after observation has already occurred. Scope does not replace the real-time experience of discomfort itself.

4-3. AI Can Infer, But It Does Not Sense

At this point, it is useful to reorganize the episode that became the starting point of this paper.

When reading the same text, the author experienced a sense that “something was off.” However, because this discomfort could not yet be verbalized, the author asked Claude and GPT separately what exactly was wrong.

Observationally, Claude detected inconsistencies in the coherence of the theoretical framework as a whole. GPT, on the other hand, pointed to localized breakdowns in inferential consistency. However, even after receiving both responses, the author’s discomfort did not disappear. Upon further examination, it became apparent that there was an additional mismatch at the linguistic and expressive level (L1), independent of the D2 points identified by either model.

At this stage, the possibility emerged that at least four different forms of D2 had been simultaneously present within the same document. This observation led to the hypothesis explored in this paper: that D2 may exist separately across different sephirotic layers.

What is important here is that Claude and GPT detected different types of D2. This may not have been merely a “difference in perspective.” Depending on each LLM’s inferential tendencies — that is, which layers of coherence it prioritizes during processing — different categories of D2 may become more easily detectable.

More importantly, however, both systems were functioning through “inference and detection,” rather than through “sensing.”

At least in current LLMs, processing does not appear to operate on the basis of a multilayer inferential coordinate system such as Scope. As a result, while LLMs may detect mismatches in layers learnable from training data — such as theoretical consistency or inferential coherence — they do not appear, at least observationally, to maintain unresolved pre-linguistic discomfort while continuing exploratory search.

In addition, LLMs possess strong completion abilities. Even when inputs contain a certain degree of ambiguity or mismatch, they tend to resolve and complete those inconsistencies contextually during generation. In other words, rather than preserving mismatches as discomfort, processing may terminate after contextual completion has already resolved the inconsistency. This is not a defect. Completion ability is one of the core strengths of LLMs. However, as a consequence, situations may arise in which “the mismatch was contextually completed, but was still actually misaligned,” and this passes through processing without being sensed as discomfort.

The exploration continued precisely because the author’s discomfort did not disappear. Had that sensing not existed, the inferential outputs produced by Claude and GPT would likely have been accepted as “sufficiently coherent explanations,” and processing would have terminated there.

Sensing functions as the starting point of inference. It is because unresolved discomfort — the sense that “something is still wrong” — continues to exist that exploration itself remains active, allowing additional D2 candidates to become targets of investigation. Conversely, if that sensing does not activate, inference may converge prematurely into what appears to be a sufficiently coherent explanation.

4-4. D2 as Undecomposed Anomaly Detection

As discussed in the previous sections, D2 does not necessarily emerge as a single isolated mismatch.

In practice, there are cases in which mismatches across multiple layers overlap simultaneously and are first sensed as an undecomposed discomfort — a vague feeling that “something is off.” What is being sensed at this stage is less like a localized error and more like a distortion in the overall structural field.

In other words, the precise location of the mismatch has not yet been decomposed. However, something within the overall coherence, connectivity, flow, or premise space is already being sensed as distorted.

For this reason, early-stage D2 sensing is not necessarily classified clearly as “an L3 mismatch” or “an L6 contradiction.” Rather, multiple anomalies may be activating simultaneously in an undecomposed state. In this paper, this condition is referred to as “undecomposed anomaly detection.”

This resembles not the detection of a single error, but the sensing that “the overall state has deviated from normal coherence.”

For example, there are cases in which correcting one D2 candidate does not eliminate the discomfort, and additional mismatches from other layers are later discovered. In such cases, the original discomfort was likely not a reaction to a single D2, but rather an undecomposed sensing of multilayer structural distortion.

What is important here is that, at this stage, even the observer themselves may be unable to explain exactly what is misaligned. The sensing exists, but classification has not yet been completed. As a result, D2 sensing is often expressed in vague forms such as “intuition,” “discomfort,” or “something feels off.” However, through post-analysis, there are cases in which multiple layers of D2 gradually become decomposed and identifiable. Furthermore, even when the undecomposed discomfort itself is similar, its subjective expression may differ depending on the observer.

This structure is also related to L8 observation. Within the framework of this paper, L8 refers to the layer responsible for whole-structure observation. Rather than detecting local logical contradictions or isolated elements, it refers to the layer in which anomalies are first observed as distortions in the overall structural field. As a result, even when the exact location of the mismatch cannot yet be verbalized, the sensing that “the structure as a whole does not align” may activate first.

Although such phenomena are often dismissed as intuition or a “sixth sense,” this is not a supernatural ability. Rather, it is closer to a condition in which multilayer information is being received simultaneously in an undecomposed state. For this reason, the process of decomposing D2 through post-analysis itself becomes a necessary exploratory process.

In this sense, D2 may emerge not merely as the detection of isolated errors, but as the undecomposed sensing of multilayer structural distortion.

The author additionally observes the possibility that undecomposed discomfort itself may exist in multiple forms. However, because both the number of observations and the precision of decomposition remain insufficient at present, this topic is not explored in detail within this paper.

Part 5. Limitations and Risks

5-1. This Is Not a Diagnostic Model

The D2 model and SCOPE presented in this paper are hypothetical frameworks for multilayered inference and observation of states and thought-generation processes occurring within dialogue and cognitive processing. They are not intended to deny or replace existing fields such as psychiatry, cognitive science, or psychology, but rather to address a different observational axis: structural estimation within the process of interaction itself.

What must be clearly stated here is that this model has fundamentally different purposes and structures from psychiatric diagnosis.

Psychiatric diagnosis is generally aimed at the classification of symptoms, the assessment of conditions, and the organization of support or treatment strategies. It operates through certain evaluative criteria, and diagnostic labels are attributed to individuals.

This model is different.

D2 is not a label indicating abnormality within a person. It is a concept for treating structural misalignments arising during dialogue and cognitive processing as observable signals. Statements such as “this person has D2” are meaningless within this framework. D2 is defined not as something a person possesses, but as something that emerges within processes of interaction and cognition.

Likewise, SCOPE is not a tool for determining whether a state is “correct” or “incorrect.” It is a coordinate system for inferring what may be occurring at different layers, and those inferences remain provisional at all times, continuously maintaining multiple possibilities simultaneously throughout the observation process.

In other words, this model does not classify people. It observes processes.

The moment this distinction collapses, SCOPE becomes misuse. The moment people begin to be treated as fixed labels, multilayer hypothesis retention disappears, and SCOPE reverses into a tool of single-interpretation fixation.

5-2. Risks of Misuse

SCOPE is an observational framework based on the retention of multiple hypotheses. However, precisely because of this structure, it also carries the risk of reversing in the opposite direction depending on how it is used.

One of the most typical forms of misuse is labeling.

Originally, SCOPE is a coordinate system for provisionally inferring what may be occurring at different layers, not for rigidly classifying human beings. However, the moment observational results become fixed into statements such as “this person is essentially a ○-layer type” or “this person always operates from ○-layer,” hypotheses begin to transform into labels.

This fixation does more than merely simplify complex cognitive states. It closes the very possibility of dialogue itself. Human states are not fixed; they fluctuate depending on context, time, relationships, and levels of cognitive or emotional load. Nevertheless, once interpretation converges into a single explanation, the observer loses sight of the possibility of change.

This does not necessarily arise from malice. The cognitive tendency to simplify complex states and compress the cost of understanding can itself induce fixation into single interpretations. Humans are not well-suited to maintaining uncertain states over long periods, and pressure naturally emerges to “settle” on an explanation. As a result, even well-intentioned observers can fall into fixation.

Furthermore, this structure can also transform into a form of controlling or dominating operation. The moment an observer becomes convinced that they have “understood” the other person, they may begin forcefully fitting the other’s reactions and states into their own interpretive model. At that point, what is occurring is no longer observation, but the imposition of a single interpretation.

Particularly in models that handle layered structures, there is a risk that observational results may be misread as hierarchies of ability, personality, or spirituality. However, the layers in this model do not represent fixed personality categories or value hierarchies. They are provisional coordinates for observing states, contexts, and processing tendencies. They are not intended to assign superiority or inferiority. Nevertheless, the moment observational results are essentialized, SCOPE loses its capacity for retaining multiple hypotheses and can reverse into a tool for reinforcing fixed worldviews and structures of domination.

SCOPE was originally designed as a structure for continuously retaining multiple possibilities. However, the moment observational results are fixed, hierarchized, and essentialized, the structure itself collapses. What remains is no longer multilayer hypothesis retention, but a fixed structure that reinforces single-interpretation lock-in.

5-3. Why Multiple-Hypothesis Retention Is Necessary

Within SCOPE, the retention of multiple hypotheses is emphasized not merely as a matter of caution or ambiguity avoidance. It is necessary because dialogue and cognitive processing themselves possess structures that cannot fundamentally be reduced to a single interpretation.

First, the possibility of misreading is always present in human observation and interpretation.

Even when observing the same statement or behavior, the underlying state or origin is not necessarily singular. What appears to be a logical statement may in some cases function as a defensive response, while what appears to be an emotional reaction may instead represent the detection of a structural inconsistency (D2). Since observers must always make inferences from limited information, immediate convergence into a single interpretation inevitably carries the risk of misrecognition.

Furthermore, even individuals themselves cannot always fully grasp their own internal states or origins. In some cases, a person may be unable to immediately verbalize why they felt a certain way, while in others they may later realize additional contributing factors. The origins of cognition, emotion, and reaction are not necessarily singular, and outputs may emerge from the interaction of multiple overlapping causes.

Second, D2 does not necessarily occur within a single layer.

In practice, within this model, there are cases in which misalignments across multiple layers overlap simultaneously and are expressed in an unresolved state. In such cases, the observed sense of inconsistency may emerge not from a single cause, but from the interference of multiple interacting factors. Nevertheless, once interpretation converges prematurely on a single explanation, other possibilities are easily overlooked.

Third, human states themselves are dynamically changing.

Cognitive states are not fixed, but fluctuate depending on context, relationships, the passage of time, fatigue, and environmental load. Even within the same individual, different processing tendencies or layer-related characteristics may become foregrounded depending on the situation. For this reason, treating observations from a single moment as permanent attributes can easily produce structural misrecognition.

This is why SCOPE requires the retention of multiple hypotheses. Multiple-hypothesis retention is not simply an act of kindness or consideration, but a necessary response to the structural complexity of the observed target itself.

SCOPE adopts multiple-hypothesis retention in order to avoid single-interpretation fixation. However, this does not eliminate the inherent incompleteness of observation itself. Rather, this model is structured to operate while explicitly presupposing that incompleteness.

5-4. The Limitations of SCOPE

SCOPE is a framework for inferring states and thought-generation processes within dialogue and cognitive processing. However, such inferences are inherently incomplete and should not be treated as definitive conclusions.

This model performs state estimation based on externally observable elements such as speech, reactions, and context. As a result, observers are always constructing hypotheses from limited information. Direct access to another person's internal state is impossible, and observational results inevitably contain margins of error and unobserved areas.

Furthermore, individual differences exist among human beings.

Even when outputs appear similar, the underlying cognitive processes or meaning structures are not necessarily the same. An interpretive model that functions effectively for one individual cannot be assumed to apply equally to another.

In addition, observation itself possesses observer dependency.

Interpretations of the same phenomenon may vary depending on the observer's assumptions, knowledge, values, experiences, and current state. Because the observer's premises and condition inevitably intervene in interpretation, complete objectivity cannot be guaranteed. In this sense, SCOPE is not a measurement device entirely independent of the observer; the observer themselves also participates as part of the structure being formed.

For this reason, the purpose of this model is not to produce “correct” definitive judgments. Rather, it is designed on the premise of maintaining multiple hypotheses while continuously retaining awareness of the limits of observation itself.

SCOPE is not a model that guarantees complete understanding. Rather, it is a structure designed to preserve the very premise that complete understanding is ultimately unattainable.

Conclusion — Toward an Era of Multilayer Observation

Through the D2 model and SCOPE, this paper has explored an approach that treats misalignment in dialogue and cognitive processing not as a singular error, but as a multilayered observational signal.

In conventional dialogue design, misalignment has often been treated as a problem to eliminate. Premise mismatches are corrected, logical inconsistencies are resolved, and emotional reactions are stabilized — such approaches have generally been prioritized. However, as demonstrated throughout this paper, misalignment does not necessarily emerge from a single cause. In some cases, it appears as the result of simultaneous interference across multiple layers. As a result, interventions that attempt to immediately eliminate misalignment risk proceeding while misidentifying the actual layer in which the phenomenon originated.

What SCOPE proposes is not the elimination of misalignment, but its observation.

Furthermore, complete alignment is not necessarily safe. Excessive pursuit of alignment can accelerate fixation into single interpretations and close down observational possibility itself. In both human relationships and interactions with LLMs, there are cases in which enforced coherence produces dependency, closure, and rigidity. D2 is not an indicator of collapse, but can instead function as an observational entry point into the internal structures underlying dialogue and cognitive processing.

In HCI design dealing with these forms of complexity, models that continuously maintain multiple possibilities may become increasingly necessary compared to models that attempt to definitively determine the “correct” interpretation.

Within this framework, SCOPE functions as an observational coordinate system, while COS functions as an operational processing structure. These are not opposing systems, but complementary structures connected through different roles.

To continuously engage with this kind of observation, isolated evaluations alone are insufficient. What is required instead are environments capable of supporting long-term, multi-observer, and dynamically interactive observation. SCOPE and COS are not theories intended to function in isolation; they become fully meaningful only when operating within Habitat-like environments capable of continuously observing and validating their behavior.

At the same time, the model presented in this paper remains a hypothetical framework originating from N=1 observation. The accumulation of multi-observer data in real environments, verification across cultures, languages, and cognitive characteristics, and the confirmation of reproducibility in interactions with LLMs — none of these have yet been fully carried out.

This model remains incomplete. However, the attempt itself — to treat misalignment as an observational signal and to describe dialogue structures while retaining multiple hypotheses — may hold observational value in the emerging era of multilayer cognition.

Not complete understanding, but the continuation of observation.

Not definitive conclusions, but the retention of multiple hypotheses.

That is the conclusion of this paper.

Appendix

This paper is part of an ongoing observational project. Related materials, continuous observation logs, and draft documents are publicly available on GitHub. Those interested in discussion, feedback, or implementation-related exploration may contact me via LinkedIn DM.

GitHub: <https://github.com/senka-l8/Layer8-Cognitive-Architecture>